Fiscal Year:	FY 2005	**Task Last Updated:**	FY 03/08/2005
PI Name:	Blawzdziewicz, Jerzy Ph.D.		
Project Title:	Drop Coalescence: Drainage and Stabilization of Thin Liquid Films		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Interfacial phenomena		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	06520-8286	**Congressional District:**	3
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Fluid Physics 01-OBPR-02: Research and Flight Experiment Opportunities
Start Date:	02/05/2002	**End Date:**	11/30/2005
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Rame, Enrique	**Contact Phone:**	216-433-2842
Contact Email:	Rame@grc.nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Rame, Enrique (National Center for Microgravity Research)		
Grant/Contract No.:	NAG3-2704		
Performance Goal No.:			
Performance Goal Text:			

Task Description:

Theory: we will develop asymptotic theories and computer simulation methods to describe the drainage and stabilization of thin liquid films.
Experiments: we will demonstrate the stabilization mechanism due to external flows. To develop theories of long-time drainage of liquid films with tangentially-mobile interfaces.

To investigate theoretically and analytically the effect of external flows on film drainage.

To examine the dynamics of film rupture when van der Waals forces are important.

To describe dynamical effects of surfactants adsorbed on film interfaces and colloidal particles or micelles present in the film. Elucidate new non-coalescence phenomenon and resolve controversies about the effect of ambient flows on film drainage dynamics.

Identify factors that control drop stability in emulsion flows.

Evaluate coalescence rates.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Task Progress:

A. Tasks achieved
1. Developed theory for multipolar expansions of Stokes flow in the space between two planar walls.

2. Developed Stokesian-dynamics algorithm for evaluation of hydrodynamic interactions between many spherical particles in a space between two planar walls.

3. Applied this algorithm to evaluate friction and mobility functions for clusters of particles moving between the walls under the action of external forces and pressure driven and shear flows.

4. Developed thermodynamic description for phase equilibria in thin liquid films stabilized by colloidal particles.

5. Using Stokesian-dynamics algorithms, evaluated effective transport properties of particle stabilized liquid films.

6. Nearly completed book chapter "Boundary integral methods for Stokes flows" for a book edited by A. Prosperetti and G. Tryggvason on computational methods for multiphase flows.

7. Conference preparation/organization: "Dynamics of Complex Fluids", at 78 ACS Colloids and Surface Science Symposium (five sessions).

8. Developed theory for analytical representation of high-frequency viscosity of emulsions of non-Newtonian fluids.

B. Publications supported by the grant:

1. J. Blawzdziewicz, V. Cristini, and M. Loewenberg, Critical Behavior of drops in linear flows. I. Phenomenological theory, Phys. Fluids 14, 2709 (2002).

2. J. Blawzdziewicz, V. Cristini, and M. Loewenberg, Multiple stationary states for deformable drops in linear Stokes flows, Phys. Fluids 15 L37 (2003).

3. J. Blawzdziewicz and S. Bhattacharya, Comment on "Drift without flux: Brownian walker with a space-dependent diffusion coefficient" by P. Lancon et al., Europhys. Lett. 63, 789 (2003).

4. V. Cristini, S. Guido, A. Alfani, J. Blawzdziewicz, and M. Loewenberg, Drop breakup and fragment size distribution in shear flow, J. Rheology 47, 1283 (2003)

5. M. B. Nemer, X. Chen, D. H. Papadopoulos, J. Blawzdziewicz, and M. Loewenberg, Hindered and enhanced coalescence of drops in Stokes flows, Phys. Rev. Lett. 92, 114501 (2004).

6. J. Blawzdziewicz, E. Wajnryb, J. A. Given, and J. B. Hubbard, Electrostatic bounds on the hydrodynamic friction of arbitrarily shaped bodies in Stokes flow, Phys. Fluids, in press.

7. S. Bhattacharya, J. Blawzdziewicz, and E. Wajnryb, Hydrodynamic interactions of spherical particles in suspensions confined between two planar walls, J. Fluid Mech., submitted.

8. J. Blawzdziewicz and E. Wajnryb, Phase equilibria in stratified thin liquid films stabilized by colloidal particles, submitted.

9. P. Vlahovska, J. Blawzdziewicz, and M. Loewenberg, Deformation of a surfactant-covered drop in linear flows, submitted.

10. S. Bhattacharya, J. Blawzdziewicz, and E. Wajnryb, Many-particle hydrodynamic interactions in parallel-wall geometry: Cartesian-representation method, Physica A, submitted.

C. Conference presentations:

1. J. Blawzdziewicz, and E. Wajnryb, Equilibrium and non-equilibrium properties of particle-stabilized thin liquid films, 78 ACS Colloid and Surface Science Symposium, June 2004, New Haven, CT.

2. S. Bhattacharya and J. Blawzdziewicz, Near-contact motion of large particles in highly-asymmetric Colloidal Mixtures, 78 ACS Colloid and Surface Science Symposium, June 2004, New Haven, CT.

3. J. Blawzdziewicz and E. Wajnryb, High-frequency linear viscoelasticity of emulsions, 78 ACS Colloid and Surface Science Symposium, June 2004, New Haven, CT.

4. Coalescence of drops in Stokes flow, M. Loewenberg, J. Blawzdziewicz, M. Loewenberg, and X. Chen, 78 ACS Colloid and Surface Science Symposium, June 2004, New Haven, CT.

	5. High-frequency linear viscosity of emulsions composed of two viscoelastic fluids, J. Blawzdziewicz and E. Wajnryb, XXI International Congress of Theoretical and Applied Mechanics, August 2004, Warsaw, Poland. 6. S. Bhattacharya, J. Blawzdziewicz, and E. Wajnryb, Hydrodynamic Interactions of Spherical Particles in Spensions Confined Between Two Planar Walls, 2004 AIChE Annual Meeting, November 2004, Austin, TX. 7. J. Blawzdziewicz and E. Wajnryb, High-frequency linear viscosity of emulsions composed of two viscoelastic fluids, 2004 AIChE Annual Meeting, November 2004, Austin, TX. 8. P. Vlahovska, and J. Blawzdziewicz, effects of surfactant on drop dynamics in linear flows, 2004 AIChE Annual Meeting, November 2004, Austin, TX.
Bibliography Type:	Description: (Last Updated: 04/23/2010)
Articles in Peer-reviewed Journals	J. Blawzdziewicz and S. Bhattacharya " Comment on “Drift without flux: Brownian walker with a space-dependent diffusion coefficient“ by P. Lancon et al." Europhys. Lett. 63, 789 (2003). , Dec-2003
Articles in Peer-reviewed Journals	V. Cristini, S. Guido, A. Alfani, J. Blawzdziewicz, and M. Loewenberg "Drop breakup and fragment size distribution in shear flow" J. Rheology 47, 1283 (2003) , Jan-2003
Articles in Peer-reviewed Journals	Nemer MB, Chen X, Papadopoulos DH, Blawzdziewicz J, Loewenberg M. "Hindered and enhanced coalescence of drops in Stokes flows." Phys Rev Lett. 2004 Mar 19;92(11):114501. PMID: 15089142 , Mar-2004
Articles in Peer-reviewed Journals	Blawzdziewicz, J; Cristini, V; and Loewenberg, M. "Multiple stationary states for deformable drops in linear Stokes flows" Phys. Fluids 15, L37 (2003) , May-2003
Articles in Peer-reviewed Journals	Blawzdziewicz J, Wajnryb E, Given J, Hubbard JB. "Sharp scalar and tensor bounds on the hydrodynamic friction and mobility of arbitrarily shaped bodies in Stokes flow." Physics of Fluids. 2005 Mar;17:033602. , Mar-2005
Presentation	J. Blawzdziewicz, and E. Wajnryb "Equilibrium and non-equilibrium properties of particle-stabilized thin liquid films" 8 ACS Colloid and Surface Science Symposium, June 2004, New Haven, CT. Jun-2004
Presentation	S. Bhattacharya and J. Blawzdziewic "Near-contact motion of large particles in highly-asymmetric Colloidal Mixtures" 78 ACS Colloid and Surface Science Symposium, June 2004, New Haven, CT. Jun-2004
Presentation	J. Blawzdziewicz and E. Wajnryb "High-frequency linear viscoelasticity of emulsions" 78 ACS Colloid and Surface Science Symposium, June 2004, New Haven, CT. Jun-2004
Presentation	M. Loewenberg, J. Blawzdziewicz, M. Loewenberg, and X. Chen "Coalescence of drops in Stokes flow" 78 ACS Colloid and Surface Science Symposium, June 2004, New Haven, CT. Jun-2004
Presentation	J. Blawzdziewicz and E. Wajnryb "High-frequency linear viscosity of emulsions composed of two viscoelastic fluids" XXI International Congress of Theoretical and Applied Mechanics, August 2004, Warsaw, Poland. Aug-2004
Presentation	S. Bhattacharya, J. Blawzdziewicz, and E. Wajnryb "Hydrodynamic Interactions of Spherical Particles in Spensions Confined Between Two Planar Walls" 2004 AIChE Annual Meeting, November 2004, Austin, TX. Nov-2004
Presentation	J. Blawzdziewicz and E. Wajnry "High-frequency linear viscosity of emulsions composed of two viscoelastic fluids" 2004 AIChE Annual Meeting, November 2004, Austin, TX. Nov-2004
Presentation	P. Vlahovska, and J. Blawzdziewicz "Effects of surfactant on drop dynamics in linear flows" 2004 AIChE Annual Meeting, November 2004, Austin, TX. Nov-2004